Yichen (Eason) Lu

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Research Scientist building real-time **voice agents** – voice interaction models and large audio language models (LALMs) – with hands-on ownership across pretraining, post-training (SFT & RL), agent interaction design, and evaluation.

Experience

Research Scientist, Anuttacon – Mountain View, CA Jan 2025 – Present

- **Voice agent post-training:** Own the post-training stage of the company's voice agent model – **hundreds of billions of parameters trained on thousands of GPUs** – driving full-cycle iteration from training recipe design to evaluation; built the post-train data pipeline and delivered **roughly 5–20% gains on audio benchmarks** (MMAU, BigBench Audio, Audio MultiChallenge), while also **strengthening the model's text reasoning and math capabilities**.
- **Agent interaction design:** Designed an **agent harness system for the voice agent** – defining how the real-time speech interaction model coordinates with the backend main agent and the user (task delegation, asynchronous result integration, proactive interjection); this harness in turn shaped the actual training-data patterns, evaluation design, and product architecture.
- **Large Audio Language Model (LALM):** Led research and development of the company's general audio understanding model (speech, audio, and music), **achieving performance comparable to state-of-the-art models** at the time; integrated it into data synthesis and filtering pipelines that underpin the voice agent's understanding capabilities.
- **Evaluation framework:** Designed and implemented an **efficient evaluation infrastructure** that runs fast, comprehensive assessments of very large models, aggregating public and internal benchmarks across understanding, generation, and dialogue; proposed new evaluation paradigms for multi-turn spoken dialogue.
- **Shipped product:** Developed the **ASR system** for *Whispers from the Star* (Anuttacon's AI-native game), supporting robust on-device speech recognition in production.

Research Assistant, CMU (advised by Prof. Shinji Watanabe) – Pittsburgh, PA July 2023 – Jan 2025

- Proposed **FastAdaSP**, a multitask-adapted token merging/pruning method for efficient Speech LM inference – published as an **EMNLP 2024 Oral**.
- Developed **SynesLM**, a unified model for audio-visual speech recognition, translation, and understanding; collaborated on audio-visual speech research published at AAAI 2025, Interspeech 2025, and SLT 2024.
- Core contributor to **ESPnet**, the open-source end-to-end speech processing toolkit; worked on discrete speech unit modeling (ICASSP 2024).

Machine Learning Engineer Intern, TrovaAI – Champaign, IL May 2023 – Aug 2023

- Designed a RAG retrieval pipeline (vector + keyword indices) over heterogeneous documents (websites, tables, and live data) for an AI agent platform.
- Built LLM-based and MT-metric (BLEU/COMET) evaluation methods for the agent system.

Machine Learning Engineer Intern, VMware, Inc. – Beijing, China July 2021 – Jan 2022

- Built an automated data analysis and reporting system for BERT-based large-scale machine translation; improved query and analysis throughput via the Elastic Stack.

Education

Carnegie Mellon University – M.S. in AI Aug 2023 – Dec 2024

- Coursework: Multimodal ML, LLM Systems, System Tool Chains for AI, Speech Recognition and Understanding

University of Illinois at Urbana-Champaign – B.S. in Statistics & Computer Science Aug 2019 – May 2023

- GPA: 3.93/4.00; Graduated with Highest Honors; Dean's List

Selected Publications

Full list: Google Scholar. (* denotes equal contribution)

[1] FastAdaSP: Multitask-Adapted Efficient Inference for Large Speech Language Model
Yichen Lu*, J. Song*, C.-H. Yang, Shinji Watanabe **EMNLP 2024 (Oral)** [Paper] [Code]

[2] ViDove: A Translation Agent System with Multimodal Context and Memory-Augmented Reasoning
Yichen Lu*, Wei Dai*, Jiaen Liu*, et al. **EMNLP 2025 System Demonstrations** [Paper] [Code]

[3] SynesLM: A Unified Approach for Audio-visual Speech Recognition and Translation
Yichen Lu*, J. Song*, X. Chang, H. Bian, S. Maiti, Shinji Watanabe **Interspeech 2024 Workshop** [Paper]

[4] Speech-DRAME: A Framework for Human-Aligned Benchmarks in Speech Role-Play
Jiatong Shi, Jionghao Han, ***Yichen Lu***, et al. **Under Review** [Paper]

[5] GALAXY: A Large-Scale Open-Domain Dataset for Multimodal Learning
Yihan Wu, ***Yichen Lu***, et al. **Interspeech 2025** [Paper]

[6] Enhancing Audiovisual Speech Recognition by Bifocal Preference Optimization
Yihan Wu, ***Yichen Lu***, Y. Peng, X. Wang, R. Song, Shinji Watanabe **AAAI 2025** [Paper]

[7] Robust Audiovisual Speech Recognition Model with Mixture-of-Experts
Yihan Wu, Yifan Peng, ***Yichen Lu***, X. Chang, R. Song, Shinji Watanabe **SLT 2024** [Paper]

[8] Exploring Speech Recognition, Translation, and Understanding with Discrete Input
Xuankai Chang, B. Yan, K. Choi, J. Jung, ***Yichen Lu***, et al. **ICASSP 2024** [Paper]

[9] Noisy Positive-Unlabeled Learning with Self-Training for Knowledge Graph Completion
Ruijie Wang, B. Li, ***Yichen Lu***, et al. **ACL 2023 Findings** [Paper]

Selected Projects

ViDove (vidove.ai), *Founder* vidove.ai | GitHub

- Built an end-to-end multimodal translation agent system for video subtitle generation with multimodal context and memory-augmented reasoning (**EMNLP 2025 System Demo**); led a 10-person engineering team and maintain the open-source repository.
- Productized the system as **vidove.ai**, an AI video translation and localization platform.
- Deployed in production with the **StarCraft 2 World Team League**, one of the largest SC2 e-sports tournaments worldwide, for tournament content translation.

Academic Service

Reviewer / Program Committee: IEEE ASRU 2025, IEEE T-ASLP, AAAI 2026, ICASSP

Skills

Languages: Python (advanced), C/C++, Rust, Java, SQL

ML & Systems: PyTorch, Torchtitan, large-scale distributed training (multi-node SFT/RL), ESPnet, LangChain, Slurm, Linux, Git, LaTeX